

Practical - 4

Definition: To analyze the provided Electric Vehicle (EV) sales data for India. This involves cleaning, transforming, and modeling the data to create an interactive Power BI dashboard. The dashboard will be used to answer key business questions regarding sales trends, market penetration, top manufacturers, regional performance, and growth rates between fiscal years 2022 and 2024.

Outcomes/Learning: Gained experience in loading and transforming multiple data sources in Power BI, creating a data model with relationships, writing DAX measures for complex calculations (like Penetration Rate and CAGR), and building a comprehensive dashboard with various visualizations to derive meaningful insights.

Required Tool: Microsoft Power BI Desktop & Power BI Query

Working: The analysis was performed by loading three CSV files (sales by state, sales by makers, and a date dimension) into Power BI. A data model was created by establishing relationships between the tables. DAX (Data Analysis Expressions) was used to create new measures for calculations not readily available in the source data. Finally, an interactive dashboard was built to visualize the findings.

Dataset Overview: The analysis is based on three distinct CSV datasets:

- **electric_vehicle_sales_by_state.csv:** This dataset contains state-level monthly sales data, detailing the number of electric vehicles sold alongside the total vehicles sold for both 2-wheeler and 4-wheeler categories.
- **electric_vehicle_sales_by_makers.csv:** This file provides a breakdown of electric vehicle sales by the manufacturer (maker), vehicle category, and date.
- **dim_date.csv:** This is a dimension table that maps each date to its corresponding fiscal year and quarter, enabling time-series analysis.

A	B	C
1	date	fiscal_yea quarter
2	01-Apr-21	2022 Q1
3	01-May-21	2022 Q1
4	01-Jun-21	2022 Q1
5	01-Jul-21	2022 Q2
6	01-Aug-21	2022 Q2
7	01-Sep-21	2022 Q2
8	01-Oct-21	2022 Q3
9	01-Nov-21	2022 Q3
10	01-Dec-21	2022 Q3
11	01-Jan-22	2022 Q4
12	01-Feb-22	2022 Q4
13	01-Mar-22	2022 Q4
14	01-Apr-22	2023 Q1
15	01-May-22	2023 Q1
16	01-Jun-22	2023 Q1
17	01-Jul-22	2023 Q2
18	01-Aug-22	2023 Q2
19	01-Sep-22	2023 Q2
20	01-Oct-22	2023 Q3
21	01-Nov-22	2023 Q3
22	01-Dec-22	2023 Q3
23	01-Jan-23	2023 Q4
24	01-Feb-23	2023 Q4
25	01-Mar-23	2023 Q4
26	01-Apr-23	2024 Q1
27	01-May-23	2024 Q1

A	B	C	D
1	date	vehicle_category	maker
2	01-Apr-21	2-Wheelers	OLA ELECTRIC
3	01-Apr-22	2-Wheelers	OKAYA EV
4	01-May-21	2-Wheelers	OLA ELECTRIC
5	01-Jun-21	2-Wheelers	OLA ELECTRIC
6	01-Jul-21	2-Wheelers	OLA ELECTRIC
7	01-Aug-21	2-Wheelers	OLA ELECTRIC
8	01-Sep-21	2-Wheelers	OLA ELECTRIC
9	01-Oct-21	2-Wheelers	OLA ELECTRIC
10	01-Nov-21	2-Wheelers	OLA ELECTRIC
11	01-Apr-21	4-Wheelers	BYD India
12	01-Apr-21	4-Wheelers	PCA Automobiles
13	01-Apr-21	4-Wheelers	BMW India
14	01-Apr-21	4-Wheelers	Volvo Auto India
15	01-Apr-21	4-Wheelers	KIA Motors
16	01-Apr-22	4-Wheelers	PCA Automobiles
17	01-Apr-22	4-Wheelers	Volvo Auto India
18	01-Apr-22	4-Wheelers	KIA Motors
19	01-May-21	4-Wheelers	BYD India
20	01-May-21	4-Wheelers	PCA Automobiles
21	01-May-21	4-Wheelers	BMW India
22	01-May-21	4-Wheelers	Mercedes-Benz AG
23	01-May-21	4-Wheelers	Volvo Auto India
24	01-May-21	4-Wheelers	KIA Motors
25	01-May-22	4-Wheelers	PCA Automobiles
26	01-May-22	4-Wheelers	Volvo Auto India
27	01-May-22	4-Wheelers	KIA Motors

A	B	C	D	E
1	date	state	vehicle_category	electric_vehides_sold
2	01-Apr-21	Sikkim	2-Wheelers	0
3	01-Apr-21	Sikkim	4-Wheelers	0
4	01-May-21	Sikkim	2-Wheelers	0
5	01-May-21	Sikkim	4-Wheelers	0
6	01-Jun-21	Sikkim	2-Wheelers	0
7	01-Jun-21	Sikkim	4-Wheelers	0
8	01-Jul-21	Sikkim	2-Wheelers	0
9	01-Jul-21	Sikkim	4-Wheelers	0
10	01-Aug-21	Sikkim	2-Wheelers	0
11	01-Aug-21	Sikkim	4-Wheelers	0
12	01-Sep-21	Sikkim	2-Wheelers	0
13	01-Sep-21	Sikkim	4-Wheelers	0
14	01-Oct-21	Sikkim	2-Wheelers	0
15	01-Oct-21	Sikkim	4-Wheelers	0
16	01-Nov-21	Sikkim	2-Wheelers	0
17	01-Nov-21	Sikkim	4-Wheelers	0
18	01-Dec-21	Sikkim	2-Wheelers	0
19	01-Dec-21	Sikkim	4-Wheelers	0
20	01-Jan-22	Sikkim	2-Wheelers	0
21	01-Jan-22	Sikkim	4-Wheelers	0
22	01-Feb-22	Sikkim	2-Wheelers	0
23	01-Feb-22	Sikkim	4-Wheelers	0
24	01-Mar-22	Sikkim	2-Wheelers	0
25	01-Mar-22	Sikkim	4-Wheelers	0
26	01-Apr-22	Sikkim	2-Wheelers	0
27	01-Apr-22	Sikkim	4-Wheelers	0

Step 1: Data Loading and Modeling

The three provided CSV datasets were imported into Power BI. Relationships were created between the tables using the 'date' column to build a star schema data model, which allows for effective filtering and analysis.

electric_vehicle_sales_by...	electric_vehicle_sales_by...	dim_date
date	date	date
Σ electric_vehicles_sold	Σ electric_vehicles_sold	Σ fiscal_year
maker	state	quarter
vehicle_category	Σ total_vehicles_sold	
Collapse ^	vehicle_category	Collapse ^

Step 2: Creating DAX Measures

Calculated measures were created using DAX to find the Penetration Rate and Compound Annual Growth Rate (CAGR) as per the formulas provided in the metadata file. These measures are essential for answering the research questions.

1 EV Penetration Rate = DIVIDE([Total Electric Vehicle Sales], [Total Vehicles Sold], 0)
1 CAGR = DIVIDE([Sales FY2024], [Sales FY2022], 0)^(1/2) - 1

Step 3: Overall EV Sales KPIs and Slicers

Key Performance Indicators (KPIs) such as Total Vehicles Sold, Total EVs Sold, and the overall Penetration Rate were added to provide a high-level summary. Slicers for 'state' and

'vehicle_category' were included to allow for interactive filtering of the report.



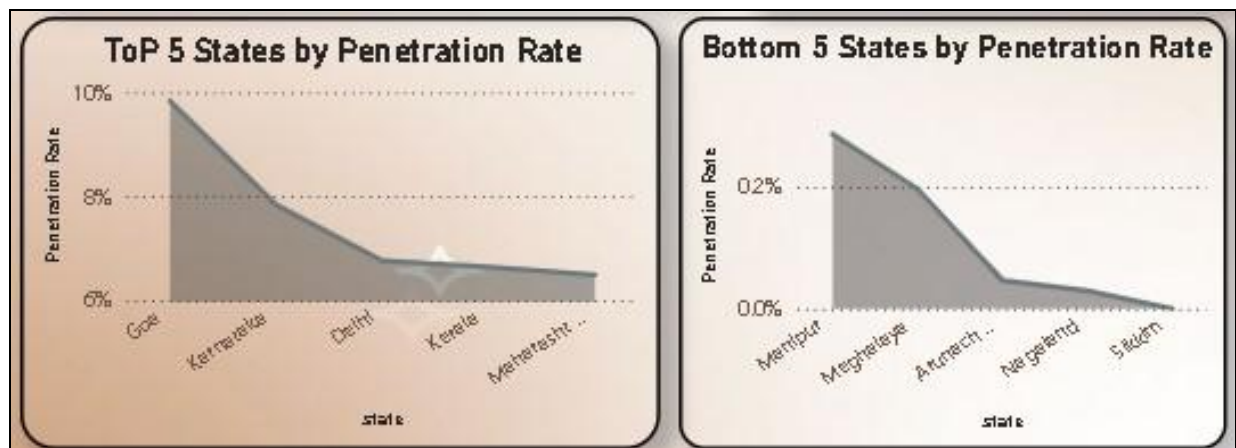
Step 4: Top & Bottom 3 Two-Wheeler Makers Analysis

Bar charts were created to visualize the top 3 and bottom 3 two-wheeler manufacturers by the number of units sold for the fiscal years 2023 and 2024, directly answering a preliminary research question.



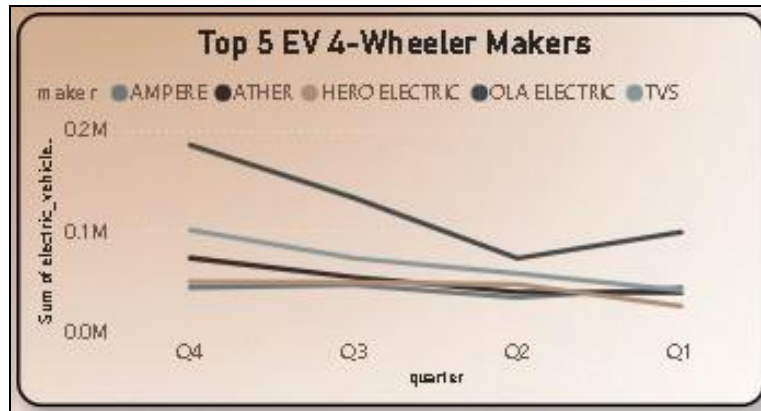
Step 5: State-wise Penetration Rate Analysis

The top 5 and bottom 5 states by EV penetration rate were identified and displayed using area charts. An area chart was used to highlight states that have shown a decline in penetration rates over the period.



Step 6: Quarterly Sales Trends and Growth Analysis

A line chart was used to show the quarterly sales trend for the top 5 EV makers. A table was used to display the Compounded Annual Growth Rate (CAGR) for different states, providing insights into market growth.



Step 7: Peak and Low Season Analysis

A chart was created to identify the peak and low season months for EV sales based on the total sales volume, helping to understand seasonal demand.

